

Question

The hexahydrate nitrate of a fifth-period metal element **M** is mixed with $\text{Cr}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ in stoichiometric ratio and gradually heated in an oxygen stream to produce compound **X** and release a reddish-brown mixed gas (Reaction 1). Further studies have shown that **X** is a paramagnetic solid with the chemical formula MCrO_4 ; 0.2786 g of solid **X** is dissolved in 0.10 mol/L H_2SO_4 without gas evolution (Reaction 2); the resulting solution is titrated to the endpoint with 23.76 mL of 0.1144 mol/L Fe^{2+} solution (Reaction 3). In the above process, the oxidation state of the metal element **M** remains unchanged.

Let the atomic number of **M** be a , the valence of Cr in **X** be b , and the absolute values of the difference between the number of reactant molecules and the number of product molecules in reactions 1, 2, and 3 (Reaction 1 is a chemical equation, 2 and 3 are ionic equations with the simplest integer ratio coefficients) be c , d , and e , respectively. Then the value of $\log_{\min\{c,d,e\}} \frac{\max\{c,d,e\}+b}{a}$ is (an error within 0.01 is considered correct):

A. All other options are incorrect

B. -0.63

C. -0.44

D. -0.37

E. -0.11

F. 0.23

G. 0.41

H. 0.64

I. 0.80

J. 1.03

K. 1.32

Answer

Correct Answer: **D**

Detailed Explanation

The number of moles of electrons transferred from 0.2786 g of solid **X** can be calculated from the titration results:

$$n(e^-) = 0.1144 \text{ mol/L} \times 23.76 \text{ mL} = 2.718 \times 10^{-3} \text{ mol}$$

CHECKPOINT

1 PTS

The number of moles of electrons transferred from 0.2786 g of solid **X** is $n(e^-) = 2.718 \times 10^{-3} \text{ mol}$

Based on reactivity, it is known that Cr will be oxidized to a valence state greater than +3. Therefore, we will discuss the valence states of Cr above +3 separately:

If it is +4, then $n(\mathbf{X}) = 2.718 \times 10^{-3} \text{ mol}$

$$M(\mathbf{X}) = 102.5 \text{ g/mol}$$

Unreasonable

If it is +6, then $n(\mathbf{X}) = 9.06 \times 10^{-4} \text{ mol}$

$$M(\mathbf{M}) = 191.5 \text{ g/mol}$$

Not a fifth-period element.

If it is +5, then $n(\mathbf{X}) = 1.359 \times 10^{-3} \text{ mol}$

$$M(\mathbf{M}) = 89.0 \text{ g/mol}$$

Is element $\mathbf{Y}$, reasonable.

CHECKPOINT

1 PTS

The valence of Cr in $\mathbf{X}$ is +5

CHECKPOINT

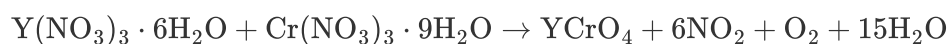
1 PTS

$\mathbf{M}$ is $\mathbf{Y}$

Therefore $a = 39, b = 5$

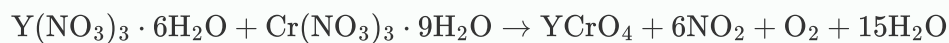
After knowing the substance, it is easy to write the equations for the three reactions:

Reaction 1:

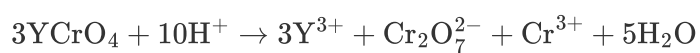


CHECKPOINT

1 PTS

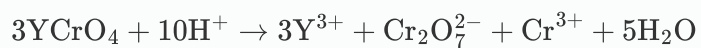


Reaction 2:

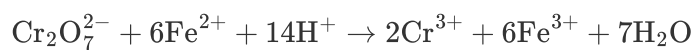


CHECKPOINT

1 PTS

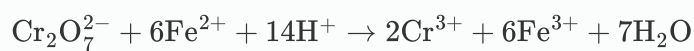


Reaction 3:



CHECKPOINT

1 PTS



Therefore $c = 21, d = 3, e = 6$

CHECKPOINT

0.2 PTS

$$a = 39$$

CHECKPOINT

0.2 PTS

$$b = 5$$

CHECKPOINT

0.2 PTS

$$c = 21$$

CHECKPOINT

0.2 PTS

$$d = 3$$

CHECKPOINT

0.2 PTS

$$e = 6$$

$$\log_{\min\{c,d,e\}} \frac{\max\{c,d,e\}+b}{a} = \log_3 \frac{21+5}{39} = -0.37$$